object appearing to be in motion to one person can appear to be at rest as viewed by another person. For example, trees on road side would appear to move backward for a person travelling in a car while the same tree would appear to be at rest for a person standing on the road side.

## 2.2 Types of Motion

In physics, motion can be classified as below.

**Linear motion:** Motion along a straight line.

**Circular motion:** Motion along a circular path.

**Oscillatory motion:** Repetitive to and fro motion of an object at regular interval of time.

**Random motion:** Motion of the object which does not fall in any of the above categories.

### 2.2.1 Uniform and Non-uniform motion

**Uniform motion**

Consider a car which covers 60 km in the first hour, 60 km in the second hour, and another 60 km in the third hour and so on. The car covers equal distance at equal interval of time. We can say that the motion of the car is uniform.

An object is said to be in uniform motion if it covers equal distances in equal intervals of time howsoever big or small these time intervals may be.

**Non-uniform motion**




Now, consider a bus starting from one stop. It proceeds slowly when it passes through crowded area of the road. Suppose, it manages to travel merely 100 m in 5 minutes due to heavy traffic and is able to travel about 2 km in 5 minutes when the road is clear. Hence, the motion of the bus is non-uniform i.e. it travels unequal distances in equal intervals of time.

Thus, an object is said to be in non-uniform motion if it covers unequal distances in equal intervals of time.

**Activity 2**

Tabulate the distance covered by a bus in a heavy traffic road in equal intervals of time and do the same for a train which is not in an accelerated motion. From your table what do you understand?

The bus covers unequal distance in equal intervals of time but the train covers equal distances in equal intervals of time.

## 2.3 Distance and Displacement

Consider a body moving from the point A. It moves along the path given in the Figure 2.1 and reaches the point B. The total length of the path travelled by the body from A to B is called distance travelled by the body. The length of the straight line AB is called displacement of the body.

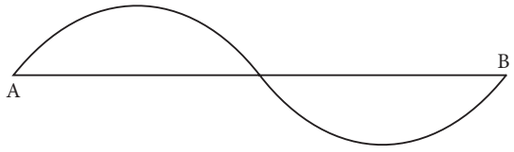



**Figure 2.1** Distance and Displacement

### 2.3.1 Distance

The actual length of the path travelled by a moving body irrespective of the direction is called the distance travelled by the body. It is measured in metre in SI system. It is a scalar quantity having magnitude only.

### 2.3.2 Displacement

It is defined as the change in position of a moving body in a particular direction. It is a vector quantity having both magnitude and direction. It is also measured in metre in SI system.



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